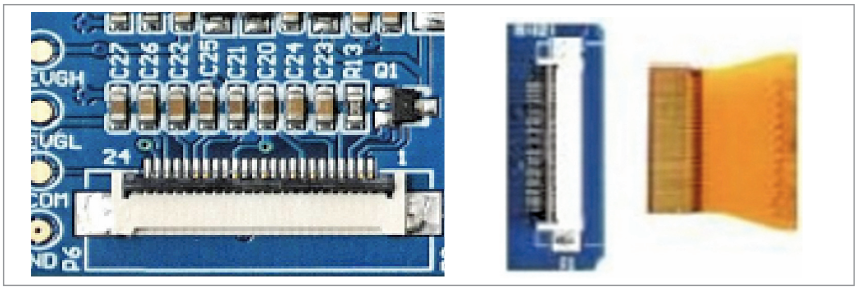

Fig. 3: FPC connector and connection with display

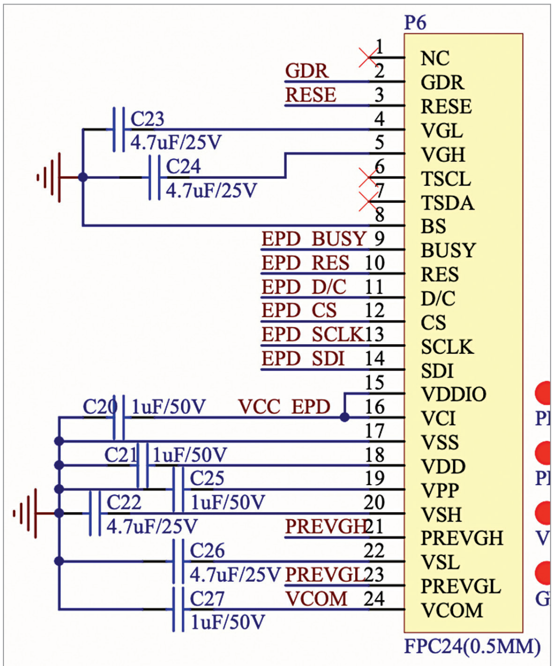

Fig. 4: EPD display connection (Source: https://v4.cecdn.yun300.cn/100001_1909185148/NFC-DEMO_SCH.pdf)

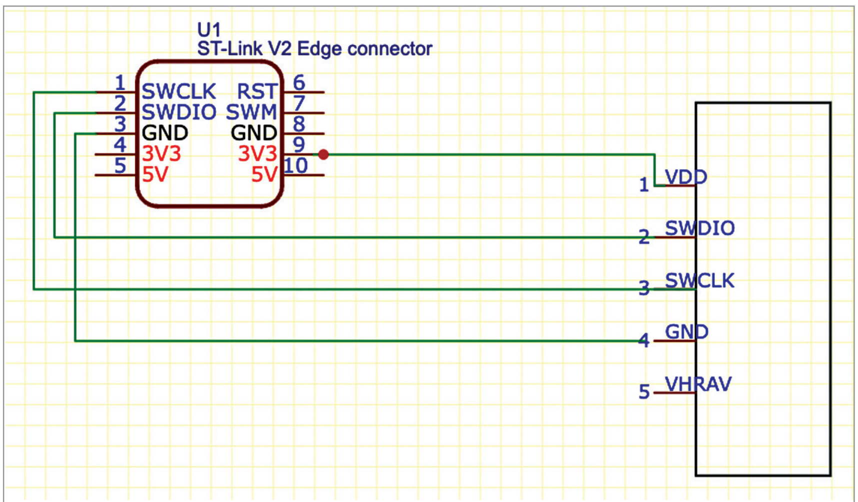

Fig. 5: Connection with the ST link emulator

Most batteryless displays on the market today utilise near-field communication (NFC). They operate by harvesting energy from NFC. To update the details on the display, one can simply use a phone with NFC or an NFC display writer device. The updated screen remains visible even when the display is not powered. Such displays are incredibly useful for batteryless sensor nodes, low-power wearable devices, price tags in clothing stores and market stalls, and marketing display screens. They can also serve as smart business cards and ID cards, allowing for updates whenever necessary.

There are numerous applications for these batteryless smart displays. Here, two versions of batteryless displays have been designed. The first version explains how NFC-based battery displays can be designed. The second version caters to those without access to PCB SMD soldering and manufacturing processes. This version describes a Wi-Fi-based batteryless display that uses a wireless coil for power during display updates. Here only the NFC-based batteryless display and Wi-Fi-based batteryless display will be delved into.

NFC-based batteryless display

If one wishes to create batteryless display identity cards, business cards, or tags without extensive SMT assembly, a modular-based design approach can be opted for. An NFC energy harvesting-based display driver module compatible with e-ink displays can be purchased. To embark on the NFC-based batteryless display, the components listed in Table 1 would be required.

However, if one prefers designing their own with PCB and SMT assembly, a reference design is available (*https://github.com/GD-SID/Hipoink*). For such a prototype (NFC-based batteryless display tag), the "DENFC-M0 Energy Harvesting" display driver and the GDEY0213B74 e-ink display have been utilised. Fig. 2 illustrates the energy harvesting display driver module.

PS: In addition to the components listed in Table 1, an Android phone and an app are also required to update the display wirelessly without using a battery. The app and all other technical details can be obtained (*https://buy-lcd.com/products/e-paper-display-spi-module-*